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# A ten-fold coordinated high-pressure structure in hafnium dihydrogen with increasing superconducting transition temperature induced by enhance pressure

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High pressure is an effective method to induce structural and electronic changes, creating novel high-pressure structures with excellent physical and chemical properties. Herein, we investigate the structural phase transition of hafnium dihydrogen (HfH<sub>2</sub>) in a pressure range of 0 GPa–500 GPa through the first-principles calculations and the crystal structure analysis by particle swarm optimization (CALYPSO) code. The high-pressure phase transition sequence of HfH<sub>2</sub> is  $I4/mmm \rightarrow Cmma \rightarrow P-3m1$  and the two phase transition pressure points are 220.21 GPa and 359.18 GPa, respectively. A newly trigonal  $P-3m1$  structure with 10-fold coordination first appears as an energy superior structure under high pressure. These three structures are all metallic with the internal ionic bonding of Hf and H atoms. Moreover, the superconducting transition temperature ( $T_c$ ) values of  $Cmma$  at 300 GPa and  $P-3m1$  at 500 GPa are 3.439 K and 19.737 K, respectively. Interestingly, the superconducting transition temperature of the  $P-3m1$  structure presents an upward trend with the pressure rising, which can be attributed to the increase of electron–phonon coupling caused by the enhanced Hf-d electronic density of states at Fermi level under high pressure.

**Keywords:** transition metal dihydrogen, first principles, phase transition, superconducting transition temperature

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## 1. Introduction

High pressure is an excellent external condition. Under high pressure, the volume of the material is compressed and the spacing between atoms or molecules in the material is gradually reduced, resulting in the overlap of electronic orbitals of adjacent atoms, resulting in the phase transition in the structure of the material. These new structures present distinctive physical and chemical properties. Kotmool *et al.*<sup>[1]</sup> predicted the phase transition of the boride RuB<sub>4</sub> under high pressure with the transition sequence of  $P6_3/mmc \rightarrow C2/c \rightarrow Immm$  and the corresponding transition pressures of 198 GPa and 388 GPa, respectively. Moreover, they also confirmed that the hardness value of the  $Immm$  phase (19.7 GPa) is much higher than that of the  $C2/c$  structure (7.6 GPa).<sup>[1]</sup> In the oxide, Dekura *et al.*<sup>[2]</sup> demonstrated that TiO<sub>2</sub> has a phase transition of cotunnite-type to Fe<sub>2</sub>P-type ( $P-62m$ ) structure at 161 GPa through the x-ray diffraction measurement. There is a reduction of band gap from 3.0 eV to 1.9 eV across the phase change, which improves the photocatalytic activity of TiO<sub>2</sub>.<sup>[2]</sup> Xin *et al.* also verified the phase transition process of  $Pm-3 \rightarrow Cm \rightarrow P1 \rightarrow I4_1md$  under high pressure in the nitride ZnN<sub>6</sub> and the  $I4_1md$  phase has an excellent energy densities of

4.70 kJ/g.<sup>[3]</sup> In the meantime, there are also numerous interesting phase transitions in other compounds under high pressure. To better understand them, our attention has focused on other compounds.

Metal hydrides exhibit peculiar electron–phonon coupling under high pressure, making them potential superconducting materials. The researches of metal hydride superconducting materials under high pressure can be mainly divided into two aspects. One is to uncover new compounds with excellent superconductivity under high pressure. Recent theoretical and experimental studies suggest that LaH<sub>10</sub>, which has a face-centered cubic  $Fm-3m$  structure, exhibits superconducting at temperatures as high as 250 K.<sup>[4–7]</sup> The critical temperatures of YLuH<sub>12</sub> can reach 283 K at 120 GPa.<sup>[8]</sup> Besides, a large number of researches exhibited that CaH<sub>6</sub>,<sup>[9]</sup> YH<sub>10</sub>,<sup>[10]</sup> and HfH<sub>10</sub>,<sup>[11]</sup> have the estimated  $T_c$  of 220 K–235 K, 303 K, and 213 K–234 K under high pressures, respectively. The other is to generate new structures through the pressure-induced phase transition for the hydrides, which can be stable at ambient pressure. It is evident that the superconductivity of these high-pressure new structures surpasses that observed under ambient pressure condition. For example, YH<sub>3</sub> crystal forms a semiconductor hcp-structured at ambient pressure, but

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the predicted  $T_c$  for the high-pressure structure fcc-YH<sub>3</sub> can achieve 40 K at 17.7 GPa.<sup>[12]</sup> Both aspects on the superconductivity of the metal hydrides aim to reveal the maximal  $T_c$  of each metal hydrides, which is urgently for investigating the  $T_c$  variation process under increasing pressure. Some structures, such as  $P6_3/mmc$ -CrH<sub>3</sub> (76 GPa–200 GPa),<sup>[13]</sup>  $P6/mmm$ -VH<sub>5</sub> (125 GPa–250 GPa),<sup>[14]</sup>  $I4/mmm$ -FeH<sub>5</sub> (150 GPa–300 GPa),<sup>[15]</sup>  $I4/mmm$ -ScH<sub>4</sub> (120 GPa–250 GPa),<sup>[16]</sup> and  $Fm-3m$ -ThH<sub>10</sub> (100 GPa–300 GPa),<sup>[17]</sup> exhibit a decreasing trend in  $T_c$  with pressure increasing. Some are the opposite (e.g.,  $R-3m$ -VH (150 GPa–250 GPa),<sup>[14]</sup>  $Immm$ -ScH<sub>8</sub> (300 GPa–400 GPa),<sup>[18]</sup> and  $Im-3m$ -MgH<sub>6</sub> (300 GPa–400 GPa)<sup>[19]</sup>). In addition, the pressure dependence of  $T_c$  of other metals is weak (e.g.,  $P6mm$ -WH<sub>5</sub> (230 GPa–300 GPa),<sup>[20]</sup>  $Fm-3m$ -YH<sub>10</sub> (250 GPa–350 GPa)<sup>[21]</sup>). The trend of  $T_c$  variation upon further compression provides guidance for achieving maximal  $T_c$  for each metal hydride, thereby motivating us to explore the  $T_c$  variation trend with the increasing pressure for the novel structures of the metal hydrides under high pressure.

Hafnium is a typical early transition metal element. Previous theoretical and experimental investigations have shown that HfH<sub>2</sub> is the only known hafnium hydride found to be stable under ambient conditions.<sup>[22–24]</sup> Through the research of the phase transition and superconducting properties of HfH<sub>2</sub> crystal at 0 GPa–300 GPa, Liu *et al.*<sup>[25]</sup> indicated that the phase transition of HfH<sub>2</sub> is  $I4/mmm \rightarrow Cmma \rightarrow P2_1/m$  at 180 GPa and 250 GPa, respectively. The  $T_c$  values of  $I4/mmm$  at 1 atm (1 atm =  $1.01325 \times 10^5$  Pa),  $Cmma$  at 180 GPa, and  $P2_1/m$  at 260 GPa are 47 mK–193 mK, 5.99 K–8.16 K, and 10.62 K–12.8 K, respectively.<sup>[25]</sup> Later on, Duda *et al.* calculated  $T_c$  values that are similar to each other.<sup>[26]</sup> However, it remains unclear whether new structures of HfH<sub>2</sub> emerge under higher pressure (> 300 GPa) and what is the trend for the superconducting transition temperature of these new structures with pressure increasing. These two questions have promoted further investigation into HfH<sub>2</sub> material at higher pressures.

In this work, we extend the research pressure range of HfH<sub>2</sub> material to 0 GPa–500 GPa and find that HfH<sub>2</sub> has a phase transition sequence of  $I4/mmm \rightarrow Cmma \rightarrow P-3m1$  at 220.21 GPa and 359.18 GPa, respectively. Among them, the fresh  $P-3m1$  structure is firstly discovered in transition metal dihydrides. The structural characteristics, dynamic stability, mechanical properties, hardness, electronic structure, bonding properties, and superconductivity of three stable structures of HfH<sub>2</sub> are systematically investigated. It is found that the coordination number of HfH<sub>2</sub> evolution process is 8→9→10 with pressure increasing. The calculated  $T_c$  of  $P-3m1$  at 500 GPa is 19.737 K. It is worthy noting that the values of  $T_c$  exhibit an upward trend with pressure further increasing.

## 2. Methods

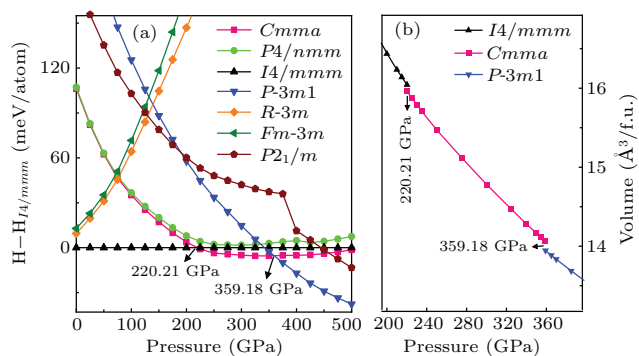
In order to extensively explore the highpressure phases of HfH<sub>2</sub>, we employed the CALYPSO code,<sup>[27,28]</sup> an unbiased structure prediction method based on the particle swarm optimization algorithm. Structure searches are conducted by using unit cells containing up to six formula units (f.u.) in the pressure range of 0 GPa–500 GPa. The superior efficiency of this method has been proved in various systems.<sup>[29–35]</sup> The follow-up structural optimizations and electronic structure calculations under different pressures were performed within density functional theory in the Vienna *ab initio* simulation package (VASP).<sup>[36]</sup> We utilized the Perdew–Burke–Ernzerhof (PBE) exchange–correlation function in generalized gradient approximation (GGA)<sup>[37,38]</sup> by using plane wave basis set and projection augmented wave (PAW)<sup>[39]</sup> method. Moreover, PAW\_PBE functional was used with valence electrons of 5p<sup>6</sup>6s<sup>2</sup>5d<sup>2</sup> and 1s<sup>1</sup> and cutoff radii of 2.5 a.u. and 0.8 a.u. to deal with Hf and H atoms, respectively. The cutoff energy of 800 eV for the plane-wave basis expansion of the electronic wave functions and a dense  $k$ -point sampling with the grid spacing less than  $2\pi \times 0.03 \text{ \AA}^{-1}$  in the Brillouin zone were selected to ensure that all the enthalpy calculations are well converged to better than 1 meV/atom. The electron localization functions (ELF)<sup>[40,41]</sup> were also computed by using VASP with ELF diagrams plotted in Visualization for Electronic and Structural Analysis (VESTA).<sup>[42]</sup> For Bader charge transfer analysis, the Bader’s quantum theory of atoms in molecules analysis method<sup>[43]</sup> was utilized. The phonon dispersion were computed by using the PHONOPY code with the supercell and finite displacement method.<sup>[44,45]</sup> The electron–phonon coupling (EPC) were computed within the linear response theory through the QUANTUM ESPRESSO package<sup>[46]</sup> with a kinetic energy cutoff of 80 Ry (1 Ry = 13.6056923 eV). In the first Brillouin zone, the  $q$ -point mesh of the electron–phonon interaction matrix elements adopted  $4 \times 4 \times 4$  for each considered structure.

## 3. Results and discussion

### 3.1. Crystal structure and structural phase transition

Firstly, based on the CALYPSO method for crystal structure prediction with 1–6 formula units (f.u.) per simulation cell, seven low-energy candidate HfH<sub>2</sub> structures ( $Cmma$ ,  $P4/nmm$ ,  $I4/mmm$ ,  $P-3m1$ ,  $R-3m$ ,  $Fm-3m$ , and  $P2_1/m$ ) are predicted over a wide pressure range of 0 GPa–500 GPa. The crystal structures and structural parameters of the above seven low-energy structures under ambient pressure are summarized in Fig. S1 and Table S1 of supplementary information. A series of the earlier known HfH<sub>2</sub> structures (e.g.,  $Fm-3m$ ,  $Cmma$ ,  $P4/nmm$ , and  $I4/mmm$ )<sup>[47–51]</sup> is successfully replicated within a specific pressure range, which demonstrates the effectiveness of the CALYPSO method adopted in structure search of HfH<sub>2</sub>. To further clarify the high-pressure

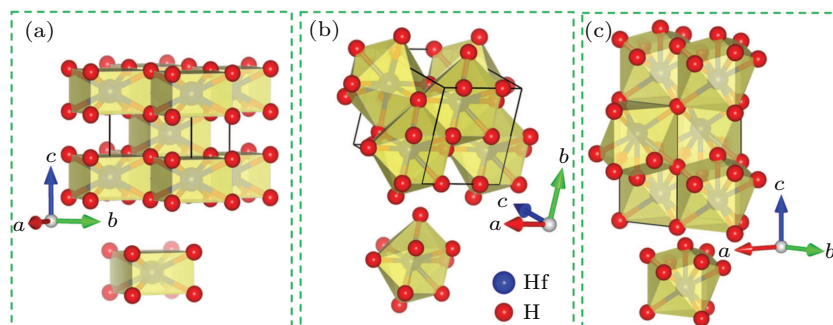
structural phase transition of  $\text{HfH}_2$ , we plot the enthalpy–pressure curve (relative to the reference phase  $I4/mmm$ ) in Fig. 1(a). As shown in Fig. 1(a), the tetragonal  $I4/mmm$  phase is the most stable structure under ambient pressure, which is also the common ground-state configuration of the group IVB of dihydrides ( $\text{HfH}_2$ ,  $\text{ZrH}_2$ , and  $\text{TiH}_2$ )<sup>[25,47,52–54]</sup> and contrary to other transitionmetal dihydrides ( $\text{VH}_2$ ,  $\text{NbH}_2$ ,  $\text{CrH}_2$ ,  $\text{CoH}_2$ )<sup>[14,49–51,55,56]</sup> whose ground-state configuration adopts the cubic  $Fm-3m$  structure. With the increase of pressure, the  $I4/mmm$  phase is no longer stable until the first phase transition occurs at 220.21 GPa. The orthorhombic  $Cmma$  phase becomes an energetically advantageous structure over the pressure range of 220.21 GPa–359.18 GPa, identical to the high-pressure phase raised by Liu *et al.*<sup>[25]</sup> Above a higher pressure of 359.18 GPa, an extraordinary trigonal  $P-3m1$  phase is the most energetically favorable and maintained at least up to 500 GPa. Hence, we can conclude that the phase transition sequence of  $\text{HfH}_2$  under high pressure is  $I4/mmm \rightarrow Cmma \rightarrow P-3m1$  with the respective transition pressures of 220.21 GPa and 359.18 GPa. This structural transformation can be further verified by the volume–pressure curve of  $\text{HfH}_2$  under the pressure range of 0 GPa–500 GPa, as displayed in Fig. 1(b). Herein, obvious volume collapses of 0.5% and 0.9% are uncovered at the transformation pressures of 220.21 GPa and 359.18 GPa, respectively. Subsequently, we focus our attention mainly on the novel  $P-3m1$  crystal to show its particularly structural characteristics, electronic structure, and superconductivity while also discussing the similar properties for the previously proposed  $I4/mmm$  and  $Cmma$  structures.



**Fig. 1.** (a) Enthalpy–pressure curves of  $\text{HfH}_2$  with respect to the  $I4/mmm$  phase and (b) volume–pressure relations for  $\text{HfH}_2$ .

### 3.2. Structural characteristics and coordination number

To elucidate the structural features and the coordination environments for the  $I4/mmm$  phase at 1 atm, the  $Cmma$  phase at 300 GPa, and the  $P-3m1$  phase at 400 GPa, we depict their structure diagrams in Fig. 2 and summarize the structural parameters in Table S2. Based on Table S2, the lattice constants of  $I4/mmm$  structure under ambient pressure are  $a = b = 3.52890 \text{ \AA}$  and  $c = 4.35240 \text{ \AA}$ , which are highly consistent with the previous results ( $a = b = 3.48130 \text{ \AA}$ ,  $c = 4.29830 \text{ \AA}$ ) derived by Liu *et al.*<sup>[25]</sup> The  $I4/mmm$  structure at atmospheric pressure is composed of an  $\text{HfH}_8$  cuboid. In this  $\text{HfH}_8$  cuboid, each Hf atom is located at the body center with eight equal Hf–H separation of  $1.71 \text{ \AA}$ , two types H–Hf–H angles of  $48^\circ$  and  $80^\circ$ , and the shortest H–H contact is  $2.17 \text{ \AA}$ . In the  $Cmma$  phase under 300 GPa, one Hf atom is surrounded by three types of H atoms with the distances of  $1.75 \text{ \AA}$ ,  $1.83 \text{ \AA}$ , and  $1.87 \text{ \AA}$ , respectively, and the H–Hf–H angles vary between  $60^\circ$  and  $90^\circ$ . Besides, the shortest H–H contact is  $1.81 \text{ \AA}$ . These results show that the coordination number of Hf increases from 8 to 9 with the formation of H-sharing  $\text{HfH}_9$  hendecahedrons. Compressed under 400 GPa, the  $P-3m1$  structure has the lattice constants:  $a = b = 2.62060 \text{ \AA}$ ,  $c = 4.77939 \text{ \AA}$ , where Hf atoms at the 2d Wyckoff sites are coordinated by three inequivalent H atoms at 2d site, 1b site, and 1a site, respectively. The Hf–H separation varies from  $1.68 \text{ \AA}$  to  $1.95 \text{ \AA}$  and the shortest H–H separation is  $1.57 \text{ \AA}$ . In addition, the H–Hf–H angle varies between  $51^\circ$  and  $125^\circ$ . The  $P-3m1$  structure consists of strongly distorted 10-fold  $\text{HfH}_{10}$  dodecahedron, which is rarely observed in other transition metal hydrides. It is noteworthy that  $\text{H}_2$  or  $\text{H}_3$  units are absent in three stable crystals as the shortest H–H bond is much longer than that of free  $\text{H}_2$  molecule ( $0.75 \text{ \AA}$ ) or  $\text{H}_3$  unit ( $< 1.00 \text{ \AA}$ ).<sup>[57–60]</sup> Moreover, the H coordination number of Hf atom becomes larger with the increase of pressure, and its growth mode is to insert an H atom at the bottom of the polyhedron in turn, which is similar to the growth mode of  $\text{TiO}_2$  and  $\text{VO}_2$ .<sup>[61–63]</sup> Interestingly, it is evident that the sharing mechanism of polyhedral small units in  $\text{HfH}_2$  varies with the coordination number. With the occurrence of  $I4/mmm \rightarrow Cmma \rightarrow P-3m1$  phase transition, it can be clearly noticed that the  $I4/mmm$  structure with 8 coordination



**Fig. 2.** Crystal structures of three stable  $\text{HfH}_2$  phases ((a)  $I4/mmm$  phase at 1 atm, (b)  $Cmma$  phase at 300 GPa, and (c)  $P-3m1$  phase at 400 GPa), together with metal coordination polyhedra, with blue and red spheres representing Hf and H atoms, respectively.

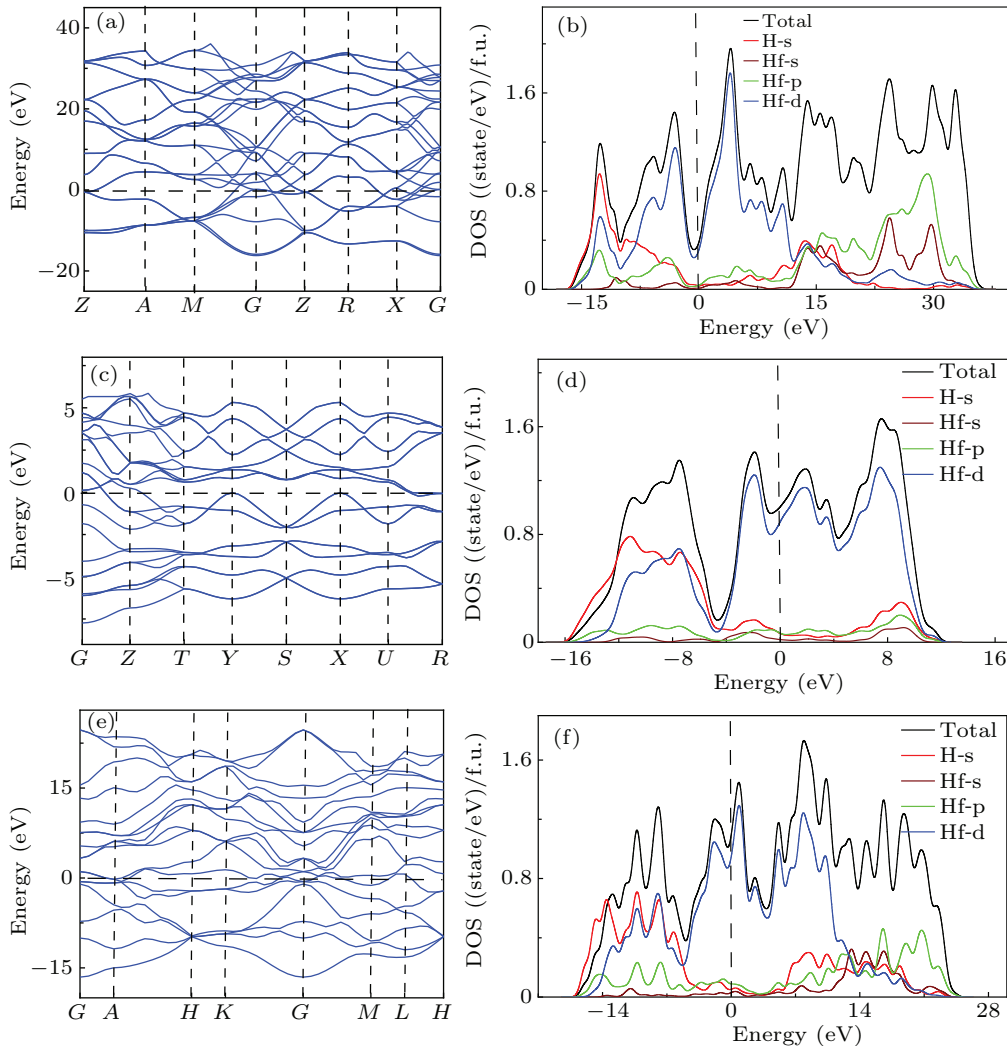


number is mainly edge sharing, and the  $Cmma$  structure with 9 coordination number is edge and surface sharing until the  $P-3m1$  structure with 10 coordination numbers is essentially transformed into surface sharing.

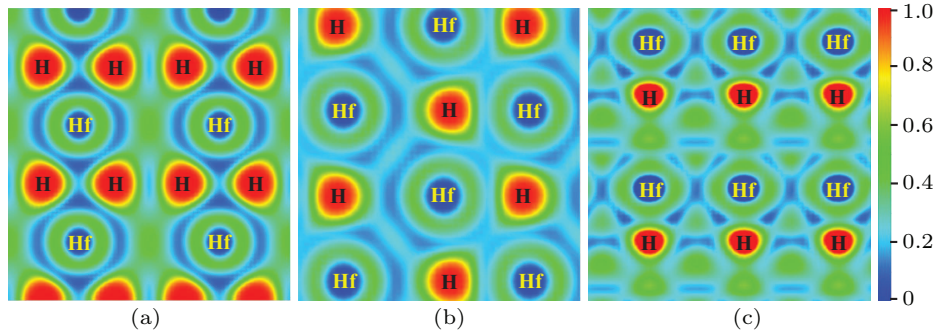
### 3.3. Electronic properties and bonding features

The electronic band structures and density of states (DOS) are calculated to investigate the electronic properties of  $I4/mmm$  at 1 atm,  $Cmma$  at 300 GPa and  $P-3m1$  at 400 GPa of  $HfH_2$  (see Fig. 3). In the meantime, considering that the spin-orbit coupling (SOC) effect of the heavy Hf atoms can induce band splitting on the metallic band structure in the conventional superconductor, we compare the electron band structures with and without the SOC effect in Fig. S2. It is found that the SOC effect has little influence on electronic band dispersion near the Fermi level. Thus, the subsequent calculations are conducted without considering SOC. As displayed in Figs. 3(a), 3(c), and 3(e), these three structures exhibit metallic characteristics because of the overlap between the conduction band and the valence band. This conclusion can also be confirmed by Figs. 3(b), 3(d), and 3(f), for there

is large total density of states at the Fermi level ( $E_F$ ). The region between two the peaks on both sides of the Fermi level is known as pseudogap and the narrow pseudogap system is corresponding to ionic bonding characteristic. It can be clearly found from Figs. 3(b), 3(d), and 3(f) that the pseudogap energy gaps of these three structures are very narrow, which reveals that  $HfH_2$  is a typical ionic bonding material. In these three stable phases of  $HfH_2$  crystal, the density of states at the Fermi level is mainly contributed by the Hf d state, which is similar to some early transition metal dihydrides ( $CrH_2$ ,<sup>[13]</sup>  $ZrH_2$ ,<sup>[47]</sup>  $VH_2$ ,<sup>[51]</sup> and  $TiH_2$ <sup>[53]</sup>). Furthermore, the  $P-3m1$  phase at 400 GPa possesses a largest Hf-d electronic density of states (0.963 (state/eV)/f.u.) at Fermi level, which is much higher than that of  $I4/mmm$  (0.323 (state/eV)/f.u.) at 1 atm and  $Cmma$  (0.880 (state/eV)/f.u.) at 300 GPa, which indicates that the phase transition significantly improves the Hf-d electronic density of states at Fermi level. Meanwhile, as exhibited in Fig. S3, the calculated Hf-d electronic DOSs of  $P-3m1$  structure at Fermi level under 360 GPa, 400 GPa, 450 GPa, and 500 GPa all present an increasing trend.



**Fig. 3.** Electronic band structures for (a)  $I4/mmm$  phase at 1 atm, (c)  $Cmma$  phase at 300 GPa and (e)  $P-3m1$  phase at 400 GPa, and total and partial density of states for (b)  $I4/mmm$  phase at 1 atm, (d)  $Cmma$  phase at 300 GPa, and (f)  $P-3m1$  phase at 400 GPa, of three stable  $HfH_2$  phases.



**Fig. 4.** The electron localization functions (ELF) of three stable HfH<sub>2</sub> phases: (a) (001) plane for the  $I4/mmm$  phase at 1 atm, (b) (100) plane for the  $Cmma$  phase at 300 GPa, and (c) (010) plane for the  $P-3m1$  phase at 400 GPa.

**Table 1.** Calculated Bader charges of Hf and H atoms in three stable HfH<sub>2</sub> phases, with  $\delta$  representing the quantity of electric charge transferred from Hf atom to H atom.

| Pressure | Space group | Atom | Charge value ( $e$ ) | $\delta$ ( $e$ ) |
|----------|-------------|------|----------------------|------------------|
| 1 atm    | $I4/mmm$    | Hf   | 8.53                 | 1.46             |
|          |             | H    | 1.73                 | -0.73            |
| 300 GPa  | $Cmma$      | Hf   | 9.09                 | 0.92             |
|          |             | H    | 1.46                 | -0.46            |
| 400 GPa  | $P-3m1$     | Hf   | 9.23                 | 0.76             |
|          |             | H    | 1.38                 | -0.38            |

To further uncover the bonding nature in three HfH<sub>2</sub> configurations, the electron localization functions (ELFs) which can accurately describe the bond type are computed, and the results are depicted in Fig. 4. Low ELF value (less than 0.5) corresponds to ionic bond or metal bond while higher ELF value (larger than 0.5) corresponds to covalent bond. As shown in Fig. 4, the ELF values around the H atom in three HfH<sub>2</sub> phases are close to 1, while those of the Hf sites are quite low. Moreover, the ELF values between Hf and the nearest H atom are smaller than 0.5, demonstrating that no electrons are localized toward the neighboring Hf–H connection. Both aspects above illustrate that the Hf–H bonds in three HfH<sub>2</sub> crystals should be ionic. In addition, the ELF value toward the neighboring H–H connection displays no electron localization, thus stating that no covalent interaction exists for neighboring H–H. This conclusion is in excellent accordance with the H–H bond length analysis. To further disclose the bonding characteristics between Hf and H atoms for three HfH<sub>2</sub> phases, the Bader mated charge analysis,<sup>[43]</sup> and the results are shown in Table 1. For three stable HfH<sub>2</sub> phases, there are large amounts of charge transferring from Hf to H atoms, which further verifies the ionic characteristic of the Hf–H bond. This prediction of the charge transferring from Hf to H atoms can be demonstrated by the element electronegativity. Based on the periodic law,<sup>[64]</sup> the electronegativity value of the Hf element (1.3) is much smaller than that of the H element (2.2), demonstrating that the charge should transfer from Hf to H atoms. This ionic bonding characteristic can also be seen in VH<sub>2</sub>,<sup>[51]</sup> TaH<sub>4</sub>,<sup>[65]</sup> FeH<sub>6</sub>,<sup>[66]</sup> YH<sub>6</sub>,<sup>[67]</sup> and CaH<sub>6</sub>.<sup>[9]</sup> However, the ionic characteristic for the H atom is contrary to that of many hydrogen-

rich compounds, such as InH<sub>3</sub>,<sup>[57]</sup> GeH<sub>4</sub>,<sup>[58]</sup> OsH<sub>6</sub>,<sup>[60]</sup> and RuH<sub>6</sub>,<sup>[68]</sup> where the H atoms are either bonded to the nearest H atoms to form an H<sub>2</sub> or H<sub>3</sub> unit, and/or covalently bonded to  $M$  ( $M = \text{In, Ge, Os, and Ru}$ ) atoms to form  $M$ –H bonds. Overall, the analysis results of DOS, ELF, and Bader all reveal the ionic characteristic between Hf and H atoms.

### 3.4. Mechanical properties and hardness

The elastic constants of the  $I4/mmm$ ,  $Cmma$ , and  $P-3m1$  structures for HfH<sub>2</sub> under corresponding pressure are calculated with the identical strain–stress means.<sup>[69]</sup> The computed consequences above are listed in Table S3. At the same time, the values of shear modulus  $G$ , bulk elastic modulus  $B$ ,  $B/G$ , Young’s modulus  $E$ , and Poisson’s ratio  $\nu$  of three stable HfH<sub>2</sub> structures can be derived from the calculated elastic constants on the basis of the Voigt–Reuss–Hill (VRH) method.<sup>[70]</sup> According to the mechanical stability criteria,<sup>[70]</sup> all three crystals satisfy mechanical stability under the corresponding pressure. Poisson’s ratio is a key physical quantity to describe the internal bonding property of the material. For the metallic material, the value  $\nu$  is usually 0.33.<sup>[71]</sup> From Table S3, the values of Poisson’s ratio  $\nu$  for the three structures are slightly larger than 0.33, which indicates that three stable HfH<sub>2</sub> phases should be metallic. In addition, to assess the values of Vickers hardness for three stable HfH<sub>2</sub> configurations, we compute their hardness values by using the empirical formula for material hardness prediction proposed by Tian *et al.*:<sup>[72]</sup>  $H_V = 0.92K^{1.137}G^{0.708}$ , where  $K = G/B$ . The calculation results are also summarized in Table S3. The hardness values of  $I4/mmm$ ,  $Cmma$ , and  $P-3m1$  structures under corresponding pressures are 1.90 GPa, 1.01 GPa, and 2.07 GPa, respectively. It should be noteworthy that the low hardness for HfH<sub>2</sub> may be related to the ionic properties between transition-metal and hydrogen atoms.<sup>[73]</sup>

### 3.5. Dynamical properties and electron–phonon coupling

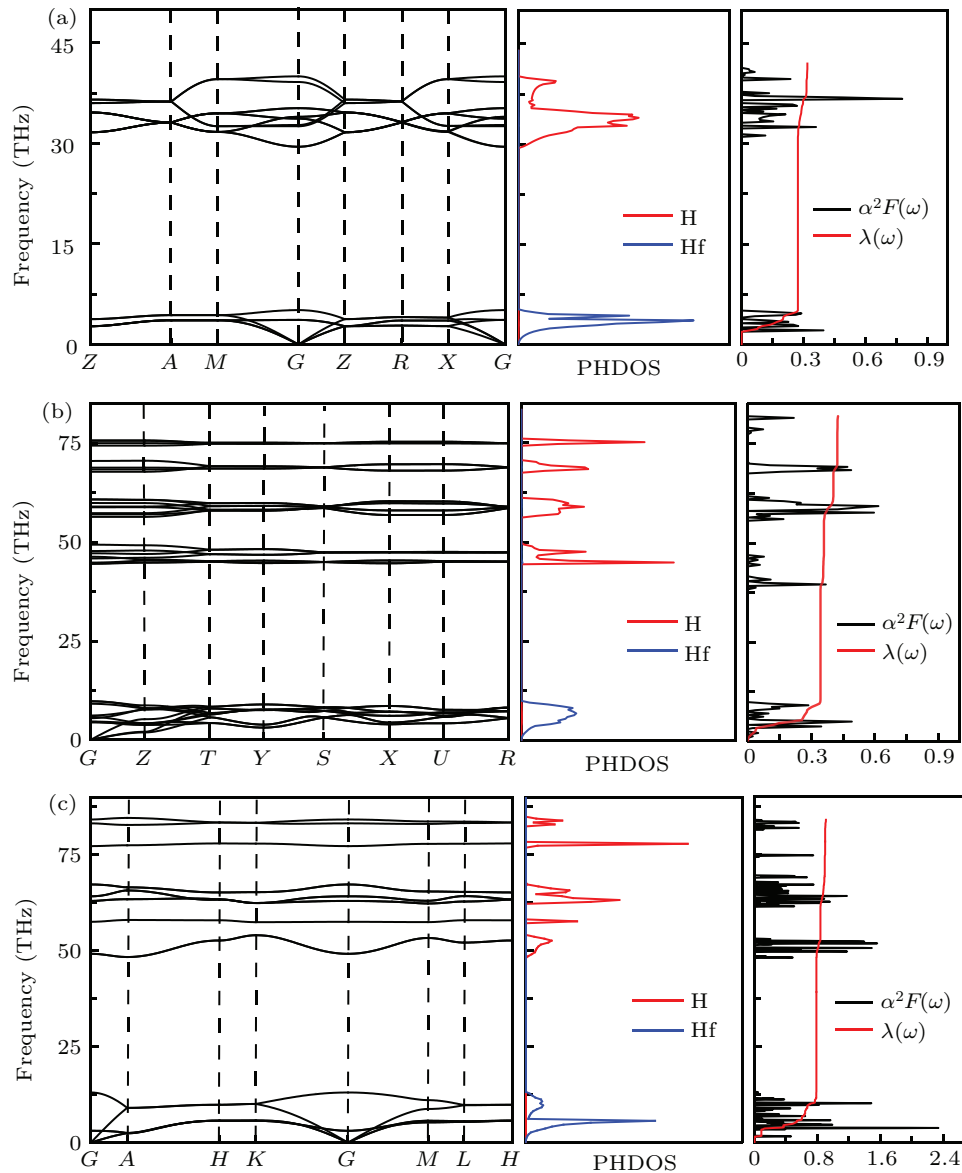
To explore the dynamical stability of  $I4/mmm$ ,  $Cmma$ , and  $P-3m1$ , we have calculated their phonon dispersion curves and the projected phonon density of states (see Fig. 5). As plotted in Fig. 5 (left and middle), the phonon dispersion

curves confirm the dynamical stabilities of these three structures, as evidenced by the absence of any imaginary frequency in the whole Brillouin zone. From Fig. 5 (left and middle), the low-frequency bands come from the vibration of the Hf

atoms while the high-frequency bands are mainly correlated with the vibration of the H atoms. This phenomenon should be ascribed to the much heavier atom mass of hafnium atom than hydrogen atom.

**Table 2.** Calculated values of electron–phonon coupling parameters  $\lambda$ , electronic density of states at the Fermi level  $N(E_F)$  (states/spin/Ry/unit cell), logarithmic average phonon frequency  $\omega_{\log}$ , and superconducting critical temperatures  $T_c$  of  $I4/mmm$ ,  $Cmma$ , and  $P-3m1$  at different pressures.

| Space group | Pressure | $\omega_{\log}$ (K) | $N(E_F)$ | $\lambda$ | $T_c$ (K) $\mu^* = 0.10$ | $T_c$ (K) $\mu^* = 0.13$ |
|-------------|----------|---------------------|----------|-----------|--------------------------|--------------------------|
| $I4/mmm$    | 1 atm    | 226.267             | 4.295    | 0.255     | 0.021                    | 0.001                    |
|             | 50 GPa   | 220.359             | 2.140    | 0.163     | 0.000                    | 0.000                    |
| $Cmma$      | 250 GPa  | 415.477             | 5.593    | 0.460     | 3.545                    | 1.940                    |
|             | 300 GPa  | 364.641             | 5.295    | 0.470     | 3.439                    | 1.936                    |
|             | 350 GPa  | 373.377             | 4.461    | 0.417     | 1.983                    | 0.935                    |
| $P-3m1$     | 360 GPa  | 273.161             | 8.333    | 0.744     | 10.957                   | 8.619                    |
|             | 400 GPa  | 306.589             | 8.248    | 0.775     | 13.459                   | 10.755                   |
|             | 450 GPa  | 280.481             | 8.223    | 0.872     | 15.532                   | 12.889                   |
|             | 500 GPa  | 309.034             | 8.259    | 0.947     | 19.737                   | 16.733                   |



**Fig. 5.** Phonon dispersion curves (left), projected phonon density of states (middle), and Eliashberg spectral function  $\alpha^2 F(\omega)$  together with electron–phonon integral  $\lambda(\omega)$  (right) of three stable  $\text{HfH}_2$  phases, for (a)  $I4/mmm$  at 1 atm, (b)  $Cmma$  at 300 GPa, and (c)  $P-3m1$  at 500 GPa.

Considering the metal behavior of three stable HfH<sub>2</sub> phases under high pressure, we calculate the values of their electron–phonon coupling (EPC) parameter  $\lambda$ , the logarithmic average phonon frequency  $\omega_{\log}$ , and electron density of states at Fermi level  $N(E_F)$  to predict their potential superconductivity. The superconducting temperature is estimated by using the Allen–Dynes modified McMillan equation to be

$$T_c = \frac{\omega_{\log}}{1.2} \exp \left[ \frac{1.04(1 + \lambda)}{\lambda - \mu^*(1 + 0.62\lambda)} \right].$$

Among them, the values of empirical constant  $\mu^*$  describing the Coulomb shielding effect are 0.10 and 0.13, respectively, as shown in Table 2. It can be found that the  $T_c$  value always decreases as the value of  $\mu^*$  increases. Although the predicted  $T_c$  of  $I4/mmm$  is only 21 mK under ambient pressure, the estimated  $T_c$  value of the  $Cmma$  phase at 300 GPa is 3.931 K and the estimated  $T_c$  of the  $P-3m1$  structure can reach 19.737 K under 500 GPa. Subsequently, we probe into the relationships between the pressure and the superconducting critical temperature  $T_c$  for three stable HfH<sub>2</sub> phases, in order to reveal the internal reasons for the change of the superconducting critical temperature  $T_c$  of HfH<sub>2</sub>. Based on Table 2, it can be clearly seen that the  $T_c$  values of  $I4/mmm$  and  $Cmma$  are negatively correlated with the pressure. On the contrary, the  $T_c$  of  $P-3m1$  presents a positive correlation with the pressure, which is relatively rare in transition metal hydrides. The trends of electron–phonon coupling parameter  $\lambda$  with the pressure intuitively lead to the trends of  $T_c$  for the three stable phases under pressure. For  $P-3m1$  phase with increased  $T_c$ , the calculated value of EPC parameter  $\lambda$  is 0.744 at 360 GPa, 0.775 at 400 GPa, 0.872 at 450 GPa, and 0.947 at 500 GPa, respectively, which emphasizes that the electron–phonon coupling strength of  $P-3m1$  is improved with the pressure increasing. To understand this coupling mechanism, the Eliashberg EPC spectral function  $\alpha^2F(\omega)$  and its integral  $\lambda$  of  $I4/mmm$  at 1 atm,  $Cmma$  at 300 GPa, and  $P-3m1$  at 500 GPa are analyzed in Fig. 5 (right). The calculation shows that the low-frequency vibrations below 5 THz from the heavy Hf atoms contribute 86% of the total  $\lambda$ , while the high-frequency vibrations (30 THz–40 THz) related to H atoms merely account for 14% of the total  $\lambda$  in  $I4/mmm$  phase. For the  $Cmma$  phase, low-frequency vibrations (below 10 THz) contribute 80%, and high-frequency vibrations (40 THz–80 THz) only contribute 20% of the total  $\lambda$ . The  $P-3m1$  phase exhibits a similar trend, where the projected phonon density of states at frequency less than 12 THz is contributed by Hf atom vibrations, which makes contribution to the most (85%) of the electron–phonon coupling coefficient  $\lambda$ , as shown in Fig. 5 (right). These results highlight that the electron–phonon coupling in three stable HfH<sub>2</sub> structures are dominated by the low-frequency Hf atom vibrations, which can also be demonstrated by the obvious Hf-d electronic den-

sity of states at Fermi level. Therefore, the significantly improved  $T_c$  value of the high-pressure novel  $P-3m1$  structure is probably due to the enhanced d-state of Hf atoms at Fermi level. Meanwhile, the increase of Hf-d electronic DOS with pressure growing can improve the electron–phonon coupling, which further leads the  $T_c$  value of  $P-3m1$  phase to increase.

#### 4. Conclusions

In summary, the stable phases of HfH<sub>2</sub> crystal within the pressure range of 0 GPa–500 GPa are determined by combining CALYPSO crystal structure prediction with first-principles calculations. The HfH<sub>2</sub> crystal transforms a structural phase transition sequence of  $I4/mmm \rightarrow Cmma \rightarrow P-3m1$  under 220.21 GPa and 359.18 GPa, respectively. Besides two known phases, namely,  $I4/mmm$  and  $Cmma$ , an unexpected trigonal  $P-3m1$  structure with 10-fold coordinated under high pressure is uncovered for the first time. Phonon dispersion curves and elastic constant calculations suggest that these three configurations are dynamically and mechanically stable. The consequences of electronic band structure and density of states show that three HfH<sub>2</sub> polymorphs have metallic properties. Further analysis on the electron localization function and Bader charge demonstrates that the HfH bond in three HfH<sub>2</sub> crystals should be ionic. In addition, the calculated superconducting critical temperature  $T_c$  of  $I4/mmm$  at 1 atm,  $Cmma$  at 300 GPa, and  $P-3m1$  at 500 GPa are 21 mK, 3.931 K, and 19.737 K, respectively. Meanwhile, the pressure dependence of the superconducting critical temperature for  $P-3m1$  phase exhibits a positive correlation trend, while  $I4/mmm$  and  $Cmma$  structures show negative correlation trends. This exceptional phenomenon of  $P-3m1$  structure is mainly responsible for the stronger electron–phonon coupling which is dominated by Hf-d orbitals.

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